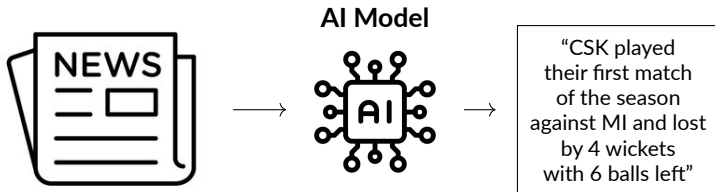


Learning to Partially Defer for Sequences

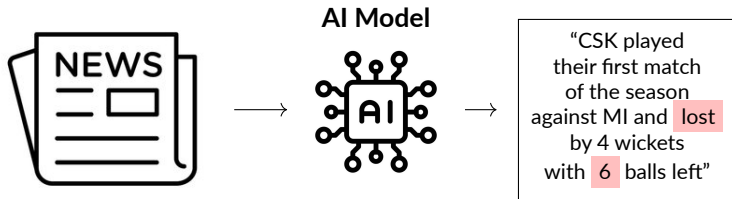
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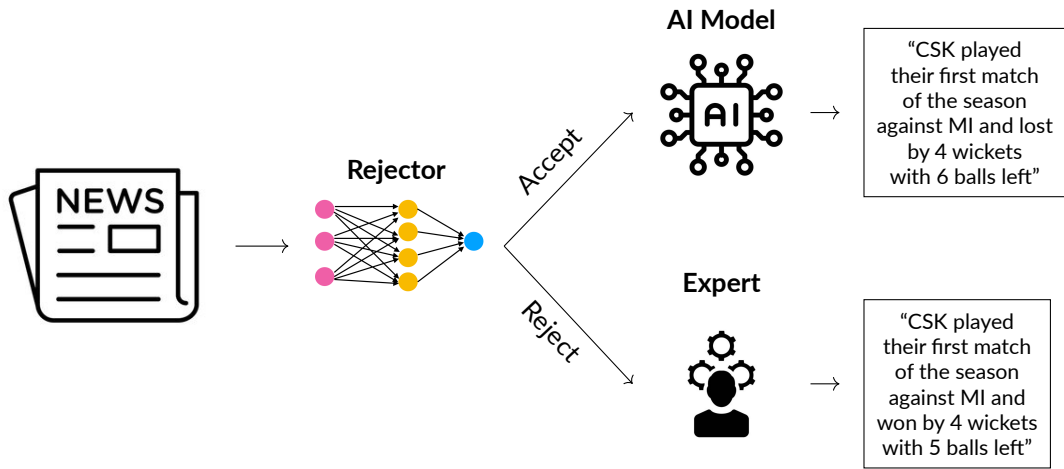
What is deferral?



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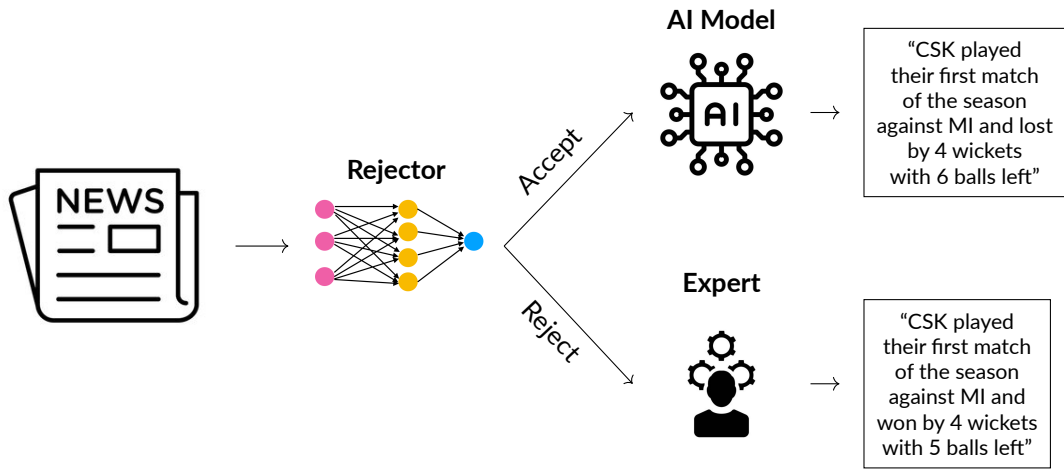


What is deferral?



There is a tradeoff between cost and accuracy

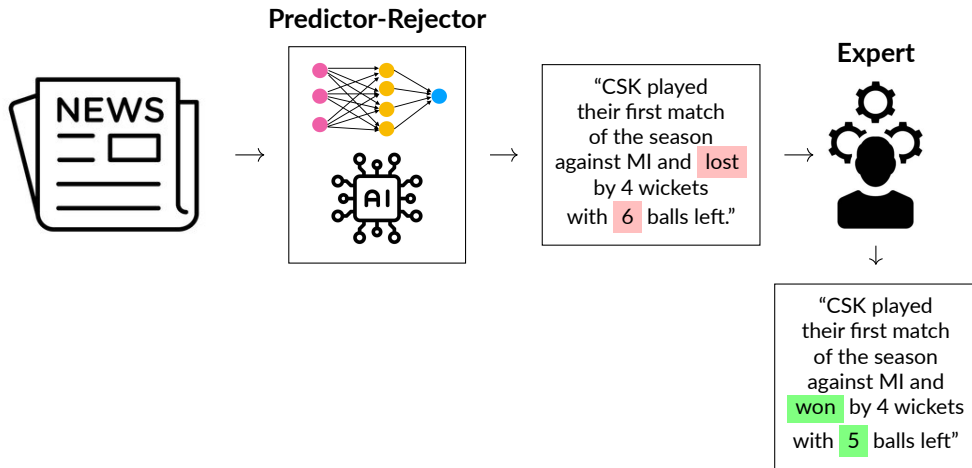
What is deferral?



Inefficiencies

- Only parts of the prediction may be incorrect
- Expert costs scale with size of output

Introducing Partial Deferral



Types of Partial Deferral

Not **ALL** experts are capable of predicting one token at a time.
We consider the following types of partial deferral:

- Token-level
- One-time

Background: Learning to Defer

Two types: confidence-based and model-based

Loss function:

$$\mathcal{L}_{\text{L2D}}(h, r, e, x, y) = l(y, h(x))\mathbb{1}_{r(x) < 0} + c(x, y, e)\mathbb{1}_{r(x) \geq 0}$$

where

- h : Predictor
- r : Rejector
- e : Expert
- $l(y, h(x))$: Model loss
- $c(x, y, e)$: Cost of deferring

Background: Learning to Defer Risk

For a fixed e and fixed h , the goal is to identify a rejector that can minimize

$$\mathcal{R}_{\text{L2D}}(r) = \mathbb{E}_{(x,y) \sim \mathcal{P}} [\mathcal{L}_{\text{L2D}}(h, r, e, x, y)]$$

Background: Consistency

Challenge: Direct minimization is difficult

Solution: Optimize a convex surrogate loss which is bayes consistent

Let

$$\mathcal{R}_l(f) = \mathbb{E}_{(x,y) \sim \mathcal{P}_{\mathcal{X},\mathcal{Y}}} [l(f(x), y)], \quad \mathcal{R}_l^* = \inf_f \mathcal{R}_l(f)$$

A surrogate loss $\phi(\cdot)$ is **Bayes consistent** if:

$$\lim_{n \rightarrow \infty} \mathcal{R}_\phi(f_n) - \mathcal{R}_\phi^* = 0 \Rightarrow \lim_{n \rightarrow \infty} \mathcal{R}_l(f_n) - \mathcal{R}_l^* = 0$$

Background: Consistency Proof

Let $\Gamma : \mathbb{R} \rightarrow \mathbb{R}$ be a nondecreasing function, continuous at 0 with $\Gamma(0) = 0$. If

$$\mathcal{R}_l(f) - \mathcal{R}_l^* \leq \Gamma(\mathcal{R}_\phi(f) - \mathcal{R}_\phi^*), \quad \forall f$$

then Bayes consistency is achieved

Token-Level Deferral: Mechanism

- $\mathcal{Y} \subseteq \mathcal{V}^L$: Output space where $\mathcal{V}$ is the vocabulary
- $\mathcal{X}$: Input space
- Expert e predicts tokens autoregressively by taking $x \in \mathcal{X}$ and $\hat{y}_{<j} = (\hat{y}_1, \dots, \hat{y}_{j-1}) \in \mathcal{V}^{j-1}$

At step j , the rejector r_j decides whether to defer:

- $r_j(x, \hat{y}_{<j}) < 0$: Use model prediction $\hat{y}_j^h = h(x, \hat{y}_{<j})$.
- $r_j(x, \hat{y}_{<j}) \geq 0$: Use expert prediction $\hat{y}_j^e$.

Token-Level Deferral: Loss Function

Token-wise Loss: At step j ,

$$\mathcal{L}_j(h, r_j, x, \hat{y}_{<j}, y) = l(y, \hat{y}_j^h) \mathbb{1}_{r_j(x, \hat{y}_{<j}) < 0} + c_j(x, \hat{y}_{<j}, y) \mathbb{1}_{r_j(x, \hat{y}_{<j}) \geq 0}$$

where $l(y, \hat{y}_j^h)$: Model loss and $c_j(x, \hat{y}_{<j}, y)$: Expert query cost.

Total Sequence Loss:

$$\mathcal{L}^{\text{Token}}(h, r, x, y) = \frac{1}{L} \sum_{j=1}^L \mathcal{L}_j(h, r_j, x, \hat{y}_{<j}, y)$$

where $r = [r_1 \ \dots \ r_L]^T$.

Risk: $\mathcal{R}_{\text{Token}}(h, r) = \mathbb{E}_{(x,y) \sim \mathcal{P}_{\mathcal{X} \times \mathcal{Y}}} [\mathcal{L}^{\text{Token}}(h, r, x, y)]$

Token-Level Deferral: Surrogate

Consider:

$$\mathcal{L}^\phi(h, r, x, y) = \frac{1}{L} \sum_{j=1}^L \mathcal{L}_j^\phi(h, r_j, x, \hat{y}_{<j}, y)$$

where

$$\mathcal{L}_j^\phi(h, r_j, x, \hat{y}_{<j}, y) = l(y, \hat{y}_j^h) \phi(r_j(x, \hat{y}_{<j})) + c_j(x, \hat{y}_{<j}, y) \phi(-r_j(x, \hat{y}_{<j}))$$

and ϕ is a convex binary surrogate loss. Ex: $\phi(x) = \log(1 + e^{-x})$

Token-Level Deferral: Bayes Consistency

Theorem 1

If $c_j(x, \hat{y}_{<j}, y) \in [\bar{c}, \bar{C}]$ for all j , $l(y, \hat{y}_j^h) \leq \bar{l}$, and ϕ be binary classification calibrated, then, for a fixed h , $\mathcal{L}^\phi$ is a Bayes consistent surrogate for $\mathcal{L}^{\text{Token}}$.

Token-Level Deferral: Bayes Consistency

Bayes optimal decision: $\text{sign}(\mathbb{P}[y = 1|x] - 1/2)$

Definition: Smallest ϕ -risk is smaller than the smallest ϕ -risk if you disobey the bayes-optimal decision.

Calibrated surrogates satisfy

$$\mathcal{R}_{\text{binary } 0-1}(f) - \mathcal{R}_{\text{binary } 0-1}^* \leq \Gamma(\mathcal{R}_{\text{binary } \phi}(f) - \mathcal{R}_{\text{binary } \phi}^*)$$

where $\mathcal{R}_{\text{binary } 0-1}$ is the binary risk and $\Gamma : \mathbb{R}^+ \rightarrow \mathbb{R}^+$ is a non-decreasing concave function.

Token-Level Deferral: Bayes Consistency

Construct a $\mathcal{X} \times \{-1, 1\}$ with mass on functions of the conditional expectations of $l(y, \hat{y}_j^h)$ and $c_j(x, \hat{y}_{<j}, y)$

By doing so,

$$\mathcal{R}_{\text{Token}}(r, h) - \mathcal{R}_{\text{Token}}^*(h) \leq \tilde{\Gamma}(\mathcal{R}_{\phi}(r, h) - \mathcal{R}_{\phi}^*(h))$$

where $\tilde{\Gamma}(z) = (\bar{l} + \bar{C}) \Gamma\left(\frac{z}{\bar{c}}\right)$, $\mathcal{R}_{\text{Token}}^*(h) := \inf_r \mathcal{R}_{\text{Token}}(r, h)$, and $\mathcal{R}_{\phi}^*(h) := \inf_r \mathcal{R}_{\phi}(r, h)$

One-Time Deferral: Mechanism

Rejector $r(x)$ selects a token index j : Expert takes over from token j onward.

Impact of Large L: Accuracy gains diminish with more deferral points

Loss Function:

$$\mathcal{L}^{\text{OneTime}}(h, r, x, y) = \sum_{j \in \mathcal{J}} (l(y, \hat{y}_{<j}) + \tilde{c}_j(x, \hat{y}_{<j}, y)) \mathbb{1}_{r(x)=j}$$

where $l(y, \hat{y}_{<j})$ is the model loss and $\tilde{c}_j(x, \hat{y}_{<j}, y)$ is the expert cost

One-Time Deferral: Surrogate

Let $\mathbf{g}$ consist of $|\mathcal{J}|$ scoring functions g_j and let $r(x) = \operatorname{argmax}_{j \in \mathcal{J}} g_j(x)$.

$$\mathcal{L}^\psi(h, r, x, y) = \sum_{j \in \mathcal{J}} (c_{\max} - l(y, \hat{y}_{<j}) - \tilde{c}_j(x, \hat{y}_{<j}, y)) \psi(\mathbf{g}(x), j)$$

where $\psi(\mathbf{g}(x), j)$ is a consistent surrogate for $\mathbb{1}_{r(x) \neq j}$, and $c_{\max} = \max_{j \in \mathcal{J}} l(y, \hat{y}_{\leq j}) + \tilde{c}_j(x, \hat{y}_{<j}, y)$.

Ex: $\psi_{ce}(\mathbf{g}(x), j) = -\log \frac{e^{g_j(x)}}{\sum_{k \in \mathcal{J}} e^{g_k(x)}}$, $\psi_{\text{mae}}(\mathbf{g}(x), j) = 1 - \frac{e^{g_j(x)}}{\sum_{k \in \mathcal{J}} e^{g_k(x)}}$

One-Time Deferral: Consistency

Theorem 3

Let $\tilde{c}_j(x, \hat{y}_{<j}, y) \leq \bar{C}$, $l(y, \hat{y}_j^h) \leq \bar{l}$, and ψ be cross entropy loss ψ_{ce} or mean absolute error ψ_{mae} . If there exists $i, j \in \mathcal{J}$ such that $|l(y, \hat{y}_i^h) + \tilde{c}_i(x, \hat{y}_{<i}, y) - l(y, \hat{y}_j^h) - \tilde{c}_i(x, \hat{y}_{<i}, y)| > \Delta > 0$, then, for a fixed h , $\mathcal{L}^\psi$ is a consistent surrogate for $\mathcal{L}^{\text{OneTime}}$.

Experiments: Setup

- **Traveling Salesman Problem (TSP):** $l(y, \hat{y}_{<j})$ = total distance of using the subtour $\hat{y}_{<j}$, $\tilde{c}_j(x, \hat{y}_{<j}, y)$ = sum of the total distance of $\hat{y}_{\geq j}^e$ and query cost α_j .
- **News Summarization (XSUM):** $l(y, \hat{y}_j^h) = \mathbb{1}_{\hat{y}_j^h \notin y}$ and $c_j(x, \hat{y}_{<j}, y) = \mathbb{1}_{e(x, \hat{y}_{<j}) \notin y}$. For one-time deferral, $l(y, \hat{y}_{\leq j}) + \tilde{c}_j(x, \hat{y}_{<j}, y)$ is jointly evaluated as the sum of 1 - ROUGE1 score and α_j
- **Weather Prediction (MWP):** $l(y, \hat{y}_j^h) = (\hat{y}_j^h - y_j)^2$ and $c_j(x, \hat{y}_{<j}, y) = (e(x, \hat{y}_{<j}) - y_j)^2$. One-time deferral: $l(y, \hat{y}_{<j}) = \|y_{<j} - \hat{y}_{<j}\|_2^2$ and $\tilde{c}_j(x, \hat{y}_{<j}, y)$ = squared error of the expert $\hat{y}_{\geq j}^e$ + query cost

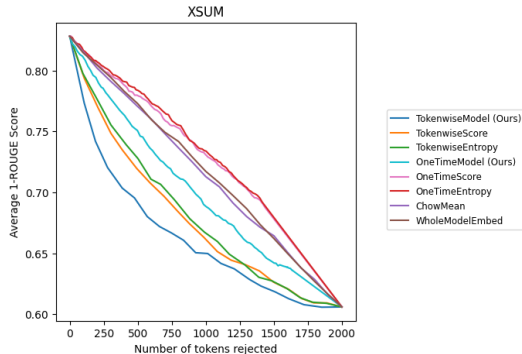
Experiments: Cost-Accuracy Tradeoff

Deferral curves: Loss against # of deferred tokens using various thresholds of $r(x)$

TSP: % increase in tour length relative to the expert tour

XSUM: $1 - \text{ROUGE1}$

MWP: $\|y - \hat{y}\|_2^2$



Cost-Loss plots.

Experiments: Cost-Accuracy Tradeoff

Method	TSP		XSUM		MWP	
	AUDC	% Improvement	AUDC	% Improvement	AUDC	% Improvement
Random	306.12(6.31)	0.00 (0.00)	1433.55 (36.98)	0.00 (0.00)	555.33 (174.75)	0.00 (0.00)
Optimal	0.00(0.00)	100.00 (0.00)	1212.00 (46.85)	15.48 (1.50)	36.66 (4.34)	93.01 (1.47)
Whole Deferrals						
ChowMean	287.01(7.73)	6.23(2.14)	1430.52 (19.09)	0.13 (3.48)	521.75 (151.71)	5.26 (9.92)
ChowSum	287.01(7.73)	6.23(2.14)	1431.41 (19.65)	0.06 (3.50)	521.75 (151.71)	5.26 (9.92)
ChowQuantile 0	301.18(9.40)	1.63(1.18)	1432.28 (25.89)	-0.01 (3.89)	528.02 (151.50)	3.91 (9.25)
ChowQuantile 0.4	287.74(5.91)	6.00(1.16)	1436.10 (24.23)	-0.27 (3.80)	528.02 (151.50)	3.91 (9.25)
ChowQuantile 0.8	288.86(9.35)	5.65(2.12)	1434.48 (19.32)	-0.15 (3.40)	528.02 (151.50)	3.91 (9.25)
ChowQuantile 1.0	295.71(6.12)	3.39(1.42)	1427.27 (18.91)	0.35 (3.50)	528.02 (151.50)	3.91 (9.25)
WholeModelEmbed	307.88(8.73)	-0.58(2.02)	1434.23 (21.57)	-0.11 (3.81)	576.24 (160.28)	-5.28 (15.09)
One-time Deferrals						
OneTimeModel	270.92(10.67)	11.52(2.41)	1398.39(22.84)	2.35 (4.03)	508.82 (143.39)	7.23 (11.80)
OneTimeScore	319.79(14.53)	-4.44(3.63)	1433.41(26.82)	-1.50 (2.13)	521.64 (134.83)	4.46 (10.14)
OneTimeEntropy	324.23(13.48)	-5.93(4.15)	1442.92(24.92)	-1.75 (2.38)	-	-
Token-level Deferrals						
TokenwiseModel	-	-	1323.92 (25.34)	7.63 (0.93)	412.45 (131.43)	25.78 (1.33)
TokenwiseScore	-	-	1354.78 (31.87)	5.48 (0.97)	707.51 (203.38)	-28.50 (6.20)
TokenwiseEntropy	-	-	1360.41 (31.50)	5.09 (0.63)	-	-

Table with Areas under Deferral Curve and % improvement over random baseline

Discussion

- Multiple Experts?
- Deferral Scheme in between token-level and one-time?
- Partial Deferral for complex outputs?

Thank you!

Appendix: Size of $\mathcal{J}$

